Pulsatile flows for Navier-Stokes fluids with energy-stable outflow boundary conditions: analysis and numerics

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1 Introduction and Notations

When carrying out numerical simulations of fluid flows in exterior domains, channels and pipes, and certain physiological geometries, artificial boundaries have to be introduced in order to define or reduce the size of computational domains and lower the computational cost of the simulations. This motivates several theoretical [7, 4, 17, 21] and numerical studies [9, 10, 15, 16] of Navier-Stokes flows with open boundary conditions.

Usually, the notation $\mathbb{T}(v,p)$ is reserved for the stress tensor, $\mathbb{T}(v,p) := 2\nu\mathbb{D}(v) - p\mathbb{I}$, where $\mathbb{D}(v)$ is the stretching tensor, $\mathbb{D}(v) = \frac{1}{2}\left(\nabla v + (\nabla v)^\top\right)$, and $\mathbb{I}$ is the identity tensor. We will use the notation $\widetilde{\mathbb{T}}(v,p) := \nu\nabla v - p\mathbb{I}$ for the so-called pseudo-stress tensor. The gradient of a vector-valued function of several variables is defined as the transpose of the Jacobian matrix, $(\nabla v)_{ij} = \frac{\partial v_j}{\partial x_i}$, $i,j = 1,...,n$.

The divergence of a tensor-valued function $\boldsymbol{T}$ is defined by $(\nabla\cdot\boldsymbol{T})_j = \frac{\partial \boldsymbol{T}_{ij}}{\partial x_i}$, $j = 1,...,n$. Notice that when $\nabla\cdot v = 0$, we have

$$\nabla\cdot\mathbb{T}(v,p) = \nu\Delta v - \nabla p = \nabla\cdot\widetilde{\mathbb{T}}(v,p).$$

In what follows, Ω is a fixed, bounded domain. However, the fluid models of the next sections may be used in fluid-structure interaction problems.

2 Directional do-nothing boundary conditions

Let $\Omega \subset \mathbb{R}^n$, $n \in \{2,3\}$, be a bounded domain with a boundary $\Gamma := \partial\Omega$ which is, at least, Lipschitz-continuous. We assume that Γ is divided into Dirichlet and Neumann portions Γ_D and Γ_N, respectively, which are disjoint relatively-open subsets of Γ with positive measure relative to Γ, $\Gamma = \overline{\Gamma_D} \cup \overline{\Gamma_N}$.

The Navier-Stokes system with classical do-nothing boundary condition (CDN) on Γ_N can be written in the form

$$\begin{cases} \partial_t v + v \cdot \nabla v = \nabla \cdot \widetilde{\mathbb{T}}(v,p) + f \text{ in } [0,T] \times \Omega \\ \nabla \cdot v = 0 \text{ in } [0,T] \times \Omega \\ v = g \text{ on } [0,T] \times \Gamma_D \\ \nu \cdot \widetilde{\mathbb{T}}(v,p) = 0 \text{ in } [0,T] \times \Gamma_N \\ v(0,x) = v_0(x),\ x \in \Omega \end{cases} \tag{2.1}$$

where ν denotes the outwards unit normal to $\partial\Omega$.

In [6, 7] (see also [4] for the case of homogeneous Dirichlet boundary conditions) a modification of the CDN condition was proposed, which leads to the following system

$$\begin{cases} \partial_t v + v \cdot \nabla v = \nabla \cdot \widetilde{\mathbb{T}}(v,p) + f \text{ in } [0,T] \times \Omega \\ \nabla \cdot v = 0 \text{ in } [0,T] \times \Omega \\ v = g \text{ on } [0,T] \times \Gamma_D \\ \nu \cdot \widetilde{\mathbb{T}}(v,p) + \dfrac{1}{2}\left[v \cdot \nu\right]^- (v - v_r) = \nu \cdot \widetilde{\mathbb{T}}(v_r,p_r) \text{ on } [0,T] \times \Gamma_N \\ v(0,x) = v_0(x),\ x \in \Omega \end{cases} \tag{2.2}$$

where

$$[y]^- := (|y| - y)/2, \quad y \in \mathbb{R},$$

and (v_r, p_r) is a suitable reference flow (which can be associated with the nonhomogeneous Dirichlet boundary conditions, as a lifting of such nonhomogeneous boundary values). The outflow boundary condition in (2.2) is called (see [4]) directional do-nothing (DDN) boundary condition. This Neumann condition takes into account the presence of the convection term in the Navier-Stokes equations and is used to prevent backflow in outflow boundary portions. In [1, 4], the authors show that the DDN condition also provides accuracy and stability for higher Reynolds numbers (in comparison with the CDN condition).

Remark 2.1. The choice of $\widetilde{\mathbb{T}}(v,p)$ is due to the compatibility of this tensor with Poiseuille flow. It is possible to replace $\widetilde{\mathbb{T}}(v,p)$ by $\mathbb{T}(v,p)$.

In [23, 24], we considered the reference flow (v_r, p_r) solution of the Stokes system

$$\begin{cases} \nabla \cdot \widetilde{\mathbb{T}}(v_r,p_r) = 0 \text{ in } \Omega \\ \nabla \cdot v_r = 0 \text{ in } \Omega \\ v_r = g \text{ on } \Gamma_D \\ \nu \cdot \widetilde{\mathbb{T}}(v_r,p_r) = 0 \text{ on } \Gamma_N. \end{cases} \tag{2.3}$$

and studied the steady problem with a regularized directional do-nothing (RDDN) bound-

ary condition

$$\begin{cases} -\nabla \cdot \widetilde{\mathbb{T}}(v,p) + v \cdot \nabla v = f \text{ in } \Omega \\ \nabla \cdot v = 0 \text{ in } \Omega \\ v = g \text{ on } \Gamma_D \\ \nu \cdot \widetilde{\mathbb{T}}(v,p) + \dfrac{1}{2}\Psi_\delta(v \cdot \nu)(v - v_r) = 0 \text{ on } \Gamma_N \end{cases} \tag{2.4}$$

where the regularized function depends on a parameter $0 < \delta \ll 1$:

$$[y]^- \approx [y]^-_\delta = \frac{1}{2}\left[y \tanh\left(\frac{y}{\delta}\right) - y + \delta\right] =: \Psi_\delta(y), y \in \mathbb{R}.$$

Of course, other regularizations of $[\cdot]^-$ are possible when defining RDDN conditions.

3 Plan of the dissertation (suggestion)

The aim is to investigate time-periodic solutions of the following problem: given time-periodic data f, g, and h, a reference flow (v_r, p_r) is first obtained using, for example, the Stokes system

$$\begin{cases} \partial_t v_r = \nabla \cdot \widetilde{\mathbb{T}}(v_r, p_r) \text{ in } [0,T] \times \Omega \\ \nabla \cdot v_r = 0 \text{ in } [0,T] \times \Omega \\ v_r = g \text{ on } \Gamma_D \\ \nu \cdot \widetilde{\mathbb{T}}(v_r, p_r) = 0 \text{ on } \Gamma_N \\ v(0,x) = v(T,x),\ x \in \Omega, \end{cases} \tag{3.1}$$

and then the corresponding velocity component v_r is used to solve the Navier-Stokes system

$$\begin{cases} \partial_t v + v \cdot \nabla v = \nabla \cdot \widetilde{\mathbb{T}}(v,p) + f \text{ in } [0,T] \times \Omega \\ \nabla \cdot v = 0 \text{ in } [0,T] \times \Omega \\ v = g \text{ on } [0,T] \times \Gamma_D \\ \nu \cdot \widetilde{\mathbb{T}}(v,p) + \dfrac{1}{2}\left[v \cdot \nu\right]^- (v - v_r) = h \text{ on } [0,T] \times \Gamma_N \\ v(0,x) = v(T,x),\ x \in \Omega. \end{cases} \tag{3.2}$$

Remark 3.1. If $g = 0$ then we do not have to solve the problem (3.1). The problem is simpler and reduces to

$$\begin{cases} \partial_t v + v \cdot \nabla v = \nabla \cdot \widetilde{\mathbb{T}}(v,p) + f \text{ in } [0,T] \times \Omega \\ \nabla \cdot v = 0 \text{ in } [0,T] \times \Omega \\ v = 0 \text{ on } [0,T] \times \Gamma_D \\ \nu \cdot \widetilde{\mathbb{T}}(v,p) + \dfrac{1}{2}\left[v \cdot \nu\right]^- v = h \text{ on } [0,T] \times \Gamma_N \\ v(0,x) = v(T,x),\ x \in \Omega. \end{cases} \tag{3.3}$$

The suggested work plan is the following:

1. Establish existence and uniqueness of weak solutions to the system (3.1)-(3.2). A possible approach consists in combining Galerkin method with a fixed point argument due to Prouse [25]. Another possible approach is a constructive existence proof using the Edelstein fixed point theorem, as in [2].

2. Develop a fully-discrete finite-differences/-elements discretization scheme, for example based on the ideas of [2].

3. Numerical simulations to confirm the advantage of using energy-stable outflow boundary conditions in the time-periodic case.

   Example 3.2. (inspired by [4], where the steady state is simulated) Consider the unit square domain
   $$\Omega := (0,1)^2,$$
   with a do-nothing boundary condition prescribed on
   $$\Gamma_N := \{0\} \times (0,1),$$
   and homogeneous Dirichlet data on
   $$\Gamma_D := \partial\Omega \setminus \Gamma_N.$$
   Also take $T = 1$. The forcing term in Ω is chosen as
   $$f(x_1, x_2, t) = \sin(2\pi t)\big(\sin(x_1) + \sin(x_2),\, 0\big),$$
   and
   $$g = 0, \quad v_r = 0, \quad h = 0.$$
   The objective is to compare the flow obtained using the CDN condition with that obtained using the DDN condition. In this example, it is not clear a priori which parts of Γ_N correspond to inflow or outflow; therefore, differences between the two types of boundary conditions are expected.

   Example 3.3. Fluid flow induced by a prescribed time-periodic inlet velocity through a backward-facing step geometry (see Fig. 1) where
   $$g_{in}(x,t) = g_{in}(x_1, x_2, t) = ((1 - \cos(2\pi t))(x_2 - 0.5)(1 - x_2), 0), \qquad f = 0.$$

   Further numerical examples can be exploited.

4. Application to Hemodynamics, with data to be provided by Jorge Tiago.

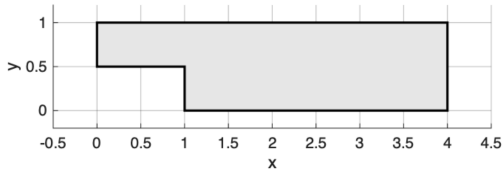

Figure 1: Backward-facing step geometry.

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